Mozhgan Hadadi

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Professional Summary

Ph.D. candidate in Mechanical Engineering at Iowa State University with experience in 3D computer vision, point cloud processing, and AI. Builds Python pipelines that turn raw 3D scans into editable 3D models using optimization, vision-language models, and LLMs, and works hands-on in field data collection, from 3D laser scanning to hyperspectral imaging. Has led the creation and public release of a 3D dataset of 1,045 scans, applied this work to crops, and published three peer-reviewed journal papers.

Work Experience

Precision Phenotyping Intern (Aug 2026 – present)

Vylor

- Building a Python pipeline that predicts lodging scores for soybean rows from 3D scan data, trained against existing ground-truth scores.
- Developing a Python tool that finds stalks, counts stems, and separates individual plants in 3D scans of maize rows, so traits can be measured for each plant instead of for the whole row.
- Validating field data during collection by reviewing data reports and visualizations to confirm data quality.

Graduate Research Assistant (Aug 2022 – present)

Computational Physics and Mechanics Laboratory, Iowa State University

- Built an automated pipeline that uses a vision-language AI model (Gemini API) to find individual leaves in 3D plant scans and turn them into editable 3D plant models. Tested on 100 field-grown maize plants, it found 99.4% of hand-labeled leaves, matched the real scans more closely than our earlier method for 99 of the 100 plants, and replaced 2–6 hours of manual tuning per plant with about 14 minutes of processing.
- Prototyped an AI agent layer (LangChain/LangGraph) on top of this pipeline, where agents ran each step, decided when to move to the next one, and ran independent steps in parallel.
- Developed FloraForge, a method in which an LLM (GPT-5) writes and refines Python 3D-modeling code through step-by-step conversation, so plant scientists without 3D-modeling experience can create editable models of new plant species. Each plant is stored as a short, human-readable settings file, and the models were checked against real 3D scans.
- Led the creation of MaizeField3D, a public dataset of 1,045 3D scans of field-grown maize from a diverse genetic panel: labeled 520 plants down to individual leaves and stalks, ran quality control, built a 3D model of every labeled plant, and packaged and released the dataset on Hugging Face and GitHub for other researchers.
- Created an open-source method that converts raw laser scans of maize into smooth, editable 3D leaf models (NURBS surfaces), and used it to produce the 3D models for all 520 labeled MaizeField3D plants. Also reconstructed plants in 3D from photos and video using NeRF-based methods.
- Ran sunlight-capture simulations on virtual maize and mung bean fields built from these 3D models to compare how much light the plants absorb across genotypes and growth stages.
- Independently set up the hyperspectral imaging system and designed the hyperspectral and 3D laser scanning data collection for pepper plants, then trained machine learning models to detect nutrient deficiency and disease.

Research & Development Intern (Summer 2017)

Automotive Industries Research & Innovation Center (AIRIC) – SAIPA – Tehran, Iran

- Researched solar vehicle technologies and sustainability metrics and prepared technical reports.
- Developed a MATLAB benchmarking tool for automotive performance comparison.

Education

- Ph.D., Mechanical Engineering, Iowa State University (ISU), 2022–present (expected Dec 2026), GPA: 4.0/4.0.
- B.Sc., Biosystem Mechanical Engineering, University of Tehran (UT), 2013–2017, GPA: 3.37/4.0.

Technical Skills

- **3D & Computer Vision:** Point cloud processing; 3D reconstruction (LiDAR, NeRF); NURBS/B-splines; optimization; procedural modeling; computer vision; light simulation (HELIOS).
- **AI & Machine Learning:** Deep learning (PyTorch); LLM and vision-language model APIs (Gemini, OpenAI, Claude); AI agents (LangChain/LangGraph); LLM-assisted code generation.
- **Data:** Dataset curation, labeling, and quality control; public data release (Hugging Face, GitHub); LiDAR and hyperspectral data collection.
- **Programming & Tools:** Python, C++, MATLAB; PyTorch, OpenCV, Open3D, NumPy/SciPy; Git, Linux; AWS (including S3); Blender, Rhino/Grasshopper, SolidWorks; COLMAP, Nerfstudio.

Honors and Awards

- Research Excellence Award, Department of Mechanical Engineering, Iowa State University (research among the top 10% at ISU), 2026.
- Selected for the Bayer Crop Science University Mentoring Program (B4U), 2025: a year-long, one-on-one mentorship with a Bayer professional.
- NSF Travel Grant, ASME IMECE 2024 Student Research Poster Competition.
- Member of the Pi Tau Sigma Honor Society.
- Member of Workforce in Ag and AI (WIAA), ISU.
- Identified as an exceptionally talented student during the undergraduate program at UT by GPA, 2017.

Leadership and Outreach

- Vice President, Iranian Students' and Scholars' Association (ISSA), ISU.
- Volunteer, Student Science and Technology Fair of Iowa (through WIAA), 2026: introduced students to drones and sensor technologies.
- President, Film and Photography Club, UT.

Publications

1- **M. Hadadi**, *et al.*, "Procedural Generation of 3D Maize Plant Architecture from LiDAR Data," *Computers and Electronics in Agriculture*, 2025. *Paper*, *GitHub Code*

2- E. Kimara*, **M. Hadadi***, *et al.*, "MaizeField3D: A Curated 3D Point Cloud and Procedural Model Dataset of Field-Grown Maize from a Diversity Panel," *Plant Phenomics*, 2025. *Paper*, *Website*, *Hugging Face* (*equal contribution)

3- **M. Hadadi**, *et al.*, "FloraForge: Procedural Generation of Editable and Analysis-Ready 3D Plant Geometric Models Using LLM-Assisted Template Design," *Smart Agricultural Technology*, 14, 102316, 2026. *Paper*, *Website*, *Code*

4- **M. Hadadi**, *et al.*, "A Vision-Language Model (VLM)-based Pipeline for End-to-End Procedural Modeling of Field-Grown Maize from Point Clouds," arXiv:2610.03468, 2026. *Paper*

Presentations

- Oral presentation, AESCAPE 2026 (USACM), session on LLMs in Computer-Aided Engineering, Dallas, TX, Sep 2026: "Automated Point-Cloud-to-Mesh Pipeline for Procedural Crop Modeling Using Vision-Language Models."
- Poster, TrAC Day (Translational AI Center), Iowa State University, Ames, IA, Apr 2026: "Automated 3D Procedural Crop Modeling Using Vision-Language Models."
- Poster, "Procedural Generation of 3D Maize Plant Architecture from LiDAR Data," presented at three conferences: ASME IMECE 2024, NSF Student Research Poster Competition, Portland, OR, Nov 2024; MLCAS 2024 Workshop, Lincoln, NE, Oct 2024; and Midwest Machine Learning Symposium (MMLS 2024), Minneapolis, MN, May 2024.